



Universidade Federal do ABC

Simulation Exercise #07: Markov Processes

INF301 – System Modeling and Simulation – W07

Prof. Luiz Henrique Bonani do Nascimento

Universidade Federal do ABC

Report Guidelines

Use a notebook on the Google Collaboratory platform to generate a report containing any explanations and comments you deem relevant, along with your codes and figures.

The graphs in the figures should be **self-explanatory**, with axis names and data captions. Use a **font size appropriate** for presentation in a document.

The language to be used is Python. However, **the use of pre-built Python libraries is not permitted**, except for those used in the examples.

Submit a **single notebook file in ipynb format**, with the file name in the format SEON_NameSurname.ipynb, where N is the SE number, Name is your first name, and Surname is your last name.

Remember that plagiarism will not be tolerated under any circumstances!

Problem 1 (40 points)

CSMA/CD (Carrier Sense Multiple-Access/Collision Detection) is a broadcast communication protocol used in local area networks. In a CSMA/CD network, a node that needs to transmit first checks the network traffic. If any traffic is detected, no message is sent (as this would cause a message error), and the medium is only retested in the next clock cycle. Conversely, if no traffic is observed, the message is sent, starting in the following clock cycle. Thus, it is possible for a free medium in one clock cycle to be occupied in the next, resulting in a collision. This behavior makes the message unreadable, and both must attempt to retransmit after waiting a random period of time.

Problem 1 (continuation)

Suppose it is known that the probability that if the medium is free at time k , it will also be free at time $k + 1$ is 0.9, and that the probability that a medium occupied at time k is also occupied at time $k + 1$ is 0.8.

- a) Obtain the state transition diagram and matrix.
- b) Obtain the theoretical steady-state matrix.
- c) Simulate this system to obtain the steady states and compare the result obtained with the result from the previous item.

Provide the proper comments on the results.

Problem 2 (60 points)

It is known that a certain car wash receives, on average, 20 cars per hour between 7:00 AM and 10:00 AM. The average drops to 8 cars per hour for the rest of the workday, which ends at 6:00 PM.

Determine analytically and using simulation:

- a) What is the average number of cars washed in a full workday?
- b) What is the probability that no more than 2 cars enter the car wash between 9:00 AM and 9:30 AM?
- c) What is the probability that more than 3 cars enter the car wash between 4:00 PM and 5:00 PM?
- d) What is the probability of having exactly 2 cars between 9:55 AM and 10:10 AM?

Provide the proper comments on the results.